CHITKARA UNIVERSITY
HIMACHAL PRADESH

# School of Engineering and Technology

**Ref. No.: CSOET/CA/INT/2026/13** **Dated: 2nd April 2026**

## NOTICE
## (Test Announcement Batch: 2024)

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|---|---|---|---|
| **Course Code : 24CA021** | | **Course Title : Data Structures** | |
| **Examination : Sessional Test II** | | **Max. Marks :** 40 | **Duration :** 90 Minutes |
| **Date of Examination :17/04/2026** | **Time : 2:00 PM** | **Mode of Examination** | **Online** |
| **Programme & Semester :** B.C.A & Semester IV | | | |

**Syllabus for Test:**

| Session No. | Topics |
|---|---|
| 1 | String Handling: Storage of Strings, Operations on Strings - Length, Concatenation |
| 2 | Substring, Insertion, Deletion, Replacement, Pattern Matching.String Applications (Palindrome, Anagram problems) |
| 3 | Linked List: Introduction, Array vs. linked list, Representation of linked lists in Memory |
| 4 | Traversing a Linked List, Insertion, Deletion, Searching into a Linked list, Type of Linked List.Problems -Reverse a Linked List-Detect Loop (Floyd's Cycle Detection – concept) Applications of Linked List |
| 5 | Stack: Array Representation of Stack, Linked List Representation of Stack, Algorithms for Push ,Pop, Peek, isEmpty and Top |
| 6 | Application of Stack: Polish Notation, Postfix Evaluation Algorithms, Infix to Postfix Conversion, Infix to Prefix Conversion. |

**Pattern of Question Paper:**

| Mark(s) | 1 Mark | 2 Marks | 5 Marks | 10 Marks |
|---|---|---|---|---|
| Total Number of questions | 10 | 5 | 2 | 1 |
| Number of questions to be attempted | 10 | 5 | 2 | 1 |

**Dr. Shankar Aggarwal**

**Name of Course Coordinator** **Signature of Course Coordinator**